

Clémence SEBE

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UNIVERSITY DEGREES

Master's Degree in Computational Biology: Analysis, Modeling, and Engineering of Biological and Medical Information 2021 - 2023
SaclAI-School Excellence Scholarship

With Honours

PARIS-SACLAY UNIVERSITY

Expertise in bioinformatics through RNA-seq, sequence and protein structure analysis, statistical and image analysis, web application development, and the design of reproducible workflows

Double Bachelor's Degree in Computer Science and Mathematics 2018 - 2021

With Honours

PARIS-SACLAY UNIVERSITY

RESEARCH EXPERIENCES

PhD Student 2023 -

LISN, PARIS-SACLAY UNIVERSITY

Thesis title: Information Extraction from Scientific Literature: A Case Study on Bioinformatics Workflows

Supervised by Sarah Cohen-Boulakia, Olivier Ferret, and Aurélie Névéol

Funded by France 2030 program ShareFAIR (Sharing reliable protocols to transform datasets into gold standards: Application to Neuro-Vascular Pathologies)

- **Developed and evaluated named entity recognition approaches** based on dictionary-based methods, encoder-based models (*BiLSTM-CRF*), and decoder-based models (*LLMs*) for named entity recognition on different corpora and entities (BioToFlow [4], CPL-Article [1], and CPL-Code [1]).
- **Extended vocabularies** and **fine-tuned** language models to improve domain-specific named entity recognition [1][4].
- Designed and coordinated an annotation campaign to **create a gold-standard dataset for entity resolution**. From two corpora: extraction of bioinformatics tool mentions from scientific publications (CPL-Article) and their corresponding executable Nextflow code (CPL-Code), and manually annotated cross-modal correspondences between tools, including unlinked cases.
- **Investigated the concordance** between bioinformatics workflow descriptions in scientific publications and their implementation in associated Nextflow code, in collaboration with the Humboldt-Universität of Berlin, during a doctoral research stay.
- **Developed expertise in the challenges of reproducibility and reusability of computational methods and research data.**
- Conducted large-scale model training and experimentation on **different high-performance computing infrastructures**, including Jean Zay HPC cluster.
- *Teaching Assistant (64h/year)* : Advanced Functional Programming, Introduction to Data Science.

Doctoral Stay

2025

HUMBOLDT-UNIVERSITÄT, BERLIN

Three-month internship within the Knowledge Management in Bioinformatics team

Supervised by Ulf Leser and funded by the project FONDA (Foundations of Workflows for Large-Scale Scientific Data Analysis)

- Worked on **entity resolution** using both supervised and unsupervised approaches, including knowledge-base bridging, semantic similarity with Sentence-BERT embeddings, and decoder-based models [1].

Research Intern

2022-2023

LISN, PARIS-SACLAY UNIVERSITY (3 and 5 months)

Knowledge Base Analysis for Bioinformatics Tools and

Creation of annotated corpora for scientific workflow from Literature and Executable Code

- **Analyzed bioinformatics tool knowledge bases** to assess their coverage, structure, and interoperability.
- **Designed and coordinated annotation campaigns** to create a richly annotated corpus of scientific publications on bioinformatics workflows (16 entity types) [1][4][8] and an annotated corpus of executable Nextflow process code [1].
 - Designed annotation schemas and developed comprehensive annotation guidelines for each annotation task.
 - Recruited, trained, and supervised 9 annotators to build high-quality annotated corpora.
 - Assessed annotation quality through inter-annotator agreement metrics and systematic resolution of annotation inconsistencies.

Study and Research Project

2021-2022

LISN, PARIS-SACLAY UNIVERSITY (9 months, 2 days per week)

Analysis of Processes in Nextflow Workflows and

Modelisation and Implementation of a Database Compiling All Information on Workflows

- Extracted structural representation of Nextflow workflow processes.
- Designed and implemented a *PostgreSQL* database integrating both high-level meta-data and process-level code information from multiple Nextflow workflows.
- Analyzed the **reuse of bioinformatics tools** across Nextflow workflows.

CO-SUPERVISION OF INTERNS

2026 Sukhjot Kaur (M1 - Paris-Saclay University) : **Development of a Web Interface for the CoPaLink Tool**, with Sarah Cohen-Boulakia, Olivier Ferret, Aurélie Névél.

2025 Anjali Badlani (M2 - University of Potsdam) : **NLP-based Workflow Analysis in Snake-make Repositories**, with Sarah Cohen-Boulakia.

PUBLICATIONS

International Journals

- [1] Clémence Sebe, Olivier Ferret, Aurélie Névél, Mahdi Esmailoghli, Ulf Leser, and Sarah Cohen-Boulakia. Supporting workflow reproducibility by linking bioinformatics tools across papers and executable code. *Bioinformatics*, 42(8):btag565, 2026

- [2] Marine Djaffardjy, George Marchment, **Sebe Clémence**, Raphaël Blanchet, Khalid Belhajjame, Alban Gaignard, Frédéric Lemoine, and Sarah Cohen-Boulakia. Developing and reusing bioinformatics data analysis pipelines using scientific workflow systems. *Computational and Structural Biotechnology Journal*, 2023

Book Chapter

- [3] Sarah Cohen-Boulakia, Frédéric Lemoine, George Marchment, Marine Djaffardjy, Alban Gaignard, **Sebe Clémence**, and Khalid Belhajjame. Reuse and reproduce bioinformatic pipelines using scientific workflow systems. In *Workflow systems for large-scale scientific data analysis*. Berlin Universities Publishing, 2026

International Peer-Reviewed Conferences

- [4] **Sebe Clémence**, Sarah Cohen-Boulakia, Olivier Ferret, and Aurélie Névéol. Extracting Information in a Low-Resource Setting: Case Study on Bioinformatics Workflows. In *Advances in Intelligent Data Analysis (IDA) XXIII*, 2025

Poster

- [5] George Marchment, Marie Schmit, **Sebe Clémence**, Frédéric Lemoine, Hervé Ménager, and Sarah Cohen-Boulakia. Representing bioinformatics Nextflow workflows in RO-Crate : challenges and opportunities. *Semantic Web Applications and Tools for Healthcare and Life Sciences (SWAT4HCLS)*, 2024
- [6] **Sebe Clémence**, Frédéric Lemoine, Alban Gaignard, Olivier Ferret, Sarah Cohen-Boulakia, and Aurélie Névéol. Towards improving workflows reproducibility : Extracting information on workflows from text and code repositories. *Intelligent Systems for Molecular Biology/European Conference on Computational Biology (ISMB/ECCB)*, 2023

National Peer-Reviewed Conferences

- [7] Marie Schmit, Melvin Selim Atay, Khalid Belhajjame, Ulysse Le Clanche, Emmanuel Coquery, Olivier Dameron, Fabien Duchateau, Alban Gaignard, Mouna El Garb, Jaffar Gura, Nicolas Lumineau, George Marchment, Camille Maumet, **Sebe Clémence**, Frédéric Lemoine, Sarah Cohen-Boulakia, and Hervé Ménager. ShareFAIR-KG, a centralised knowledge base of scientific workflows. In *Journées Ouvertes en Biologie, Informatique et Mathématiques (JOBIM)*, 2026
- [8] **Sebe Clémence**, Sarah Cohen-Boulakia, Olivier Ferret, and Aurélie Névéol. Extraction d'entités nommées décrivant des chaînes de traitement bioinformatiques dans des articles scientifiques en anglais. In *Traitement Automatique des Langues Naturelles (TALN)*, 2024
- [9] **Sebe Clémence**, Aurélie Névéol, Sarah Cohen-Boulakia, and Alban Gaignard. Extraction d'informations sur les workflows scientifiques à partir de la littérature. In *Extraction et Gestion des Connaissances (EGC), RNTI-E-39*, 2023

DEVELOPED TOOLS AND RESOURCES

CoPaLink

CoPaLink is a **tool** designed to **automatically extract the names of bioinformatics tools** mentioned in a scientific article describing a Nextflow workflow, as well as in the executable source code of the workflow. A **linking step** between the bioinformatics tool names from both sources is then performed.

GitLab repository: <https://gitlab.liris.cnrs.fr/sharefair/copalink>

CPL-Article

CPL-Article is an **annotated corpus** describing **bioinformatics tool names in English publications** describing bioinformatics workflows. This corpus is composed of 50 articles related to Nextflow workflows randomly selected from PubMed Central with a total of 76,002 tokens and 1,523 annotated tokens corresponding to 1,455 tool occurrences (425 unique tools).
Zenodo link: <https://doi.org/10.5281/zenodo.20746904>

CPL-Code

CPL-Code is an **annotated corpus** describing **bioinformatics tool names in executable code of Nextflow workflows**. This corpus comprises 1,368 Nextflow workflow processes, randomly selected from GitHub, totaling 137,337 tokens and 3,401 annotated tokens, corresponding to 3,398 tool occurrences (537 unique tools).
Zenodo link: <https://doi.org/10.5281/zenodo.20746970>

CPL-Gold-Entity-Resolution

CPL-Gold-Entity-Resolution is an **annotated entity resolution corpus** between publications and executable code for Nextflow workflows. This corpus comprises 39 workflows extracted from CPL-Article and CPL-Code. Bioinformatics tool mentions were extracted from both corpora, and cross-modal correspondences between tools were manually annotated, including unlinked cases. 490 cross-model links are represented.
Zenodo link: <https://doi.org/10.5281/zenodo.20746994>

BioToFlow

BioToFlow is an **annotated corpus** describing **bioinformatics workflows in English publications**. This corpus is composed of 52 articles (26 articles related to Nextflow workflows and 26 on Snakemake workflows, randomly selected from PubMed) with a total of 78,419 tokens, 27,786 annotated tokens with 16 entities.
Zenodo link: <https://doi.org/10.5281/zenodo.14900544>

COMPUTER SKILLS

Programming Languages	:	PYTHON, SQL, BASH, C/C++, JAVA, OCAML
Machine Learning	:	PyTorch, Hugging Face Transformers, scikit-learn
Workflow Management System	:	Nextflow
Containerization	:	Docker
Databases	:	PostgreSQL
Other Tools	:	L ^A T _E X, Git, R

OTHER RESEARCH-RELATED ACTIVITIES

Oral Presentation at JOBIM (Open Days in Biology, Computer Science, and Mathematics)	2026
<i>Presented research on linking bioinformatics tools described in scientific papers with their executable code : CoPaLink tool [1]</i>	
PhD Student Representative at LISN	2024-2025
<i>Participation in the laboratory council, organization of events</i>	

(Invited) Oral Presentation at the Scientific Day of the Bioinformatics Axis of Biogenouest 2024

Title: Information Extraction on Scientific Workflows from English-Language Literature: Preliminary Results, May 24, Rennes

Oral Presentation during *ShareFAIR* Project Days 2023 - 2026

- *Extracting and Linking Workflow Information between Bioinformatics Literature and Source Code*, December 10-11, 2025
- *BioToFlow corpus*, November 28-29, 2024
- *Extraction of Bioinformatics Tool Names from the Scientific Literature*, November 20-21, 2023

Oral Presentation during Team Seminars at LISN 2023-2026

- *Bioinformatics Team*, January 12, 2023, and February 5, 2026
- *Reading Group of the Language Sciences and Technologies Department*, September 27, 2024

Oral Presentation at the Junior Multidisciplinary Congress of Paris-Saclay University 2022-2023

- *Information Extraction of Scientific Workflows from English Articles*, June 8, 2023
- *Reusing Bioinformatics Analysis Workflows*, June 2, 2022

Oral Presentation at R2P2 Project Days (Multimodal Data Analysis for Complex Diseases Through the Design and Implementation of Reproducible and Reusable Protocols) 2021

OUTREACH ACTIVITIES

Film-Debate 2026

Participation in a Film-Debate with high school students from Lycée Blaise Pascal, Orsay on April 2. Film: "Le Théorème de Marguerite"

Speed-Meeting 2026

- *Ambition: Science for Women*, mentoring program with high school girls from Lycée Marceau in Chartres, March 12
- *Career guidance for gender equality*, with 9th-grade students from Collège Charles Péguy, Palaiseau, December 15